

2) - Variables

$$2^4 = 16$$

$f(a, b, c, d)$

$$f = \bar{a}\bar{b}\bar{c} + ab\bar{c} + \bar{d}$$

	$\bar{c}\bar{d}$	$\bar{c}d$	cd	$c\bar{d}$
$\bar{a}\bar{b}$	1	1		1
$\bar{a}b$	1			1
$a\bar{b}$	1	1		1
ab	1			1

1100

4- Variables $2^4 = 16$

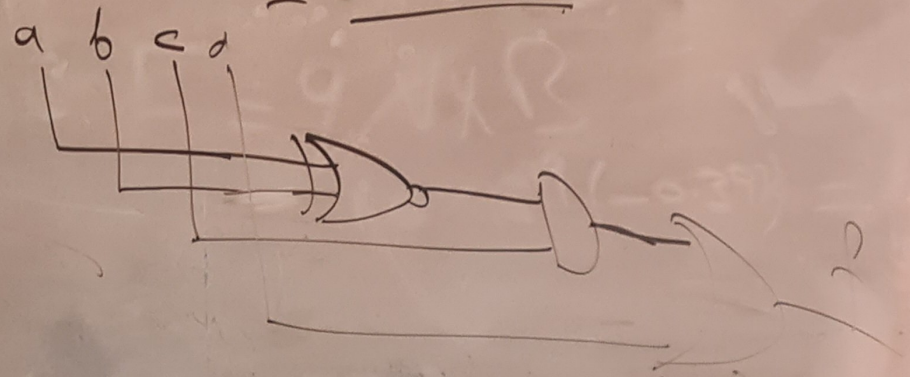
$f(a,b,c,d)$

$$f = \underline{\bar{a}\bar{b}\bar{c} + ab\bar{c}} + \bar{d}$$

	$\bar{c}\bar{d}$	$\bar{c}d$	cd	$c\bar{d}$
$\bar{a}\bar{b}$	1	1		
$\bar{a}b$	1			1
$a\bar{b}$		1	1	1
ab		1		1

$$f = \bar{c}(\bar{a}\bar{b} + ab) + \bar{d}$$

$$= \bar{c}(\overline{a \oplus b}) + \bar{d}$$



1100

2) - Variables

$$f = \bar{x}v + \bar{y}v$$

00

f =

	$\bar{z}\bar{v}$	$\bar{z}v$	zv	$z\bar{v}$
$\bar{x}\bar{y}$		1	1	
$\bar{x}y$		1	1	
$x\bar{y}$				
xy		1	1	

2) - Variables

$$f = \bar{x}v + \bar{y}v$$

	$\bar{z}\bar{v}$	$\bar{z}v$	zv	$z\bar{v}$
$\bar{x}\bar{y}$		1	1	
$\bar{x}y$		1	1	
xy				
$x\bar{y}$		1	1	

00
03

$$f =$$

$$\sum \prod$$

4- Variables

Product of - Sum Simplification

	$\bar{z}\bar{v}$	$\bar{z}v$	zv	$z\bar{v}$
$\bar{x}\bar{y}$	1	0	0	1
$\bar{x}y$	1	0	0	1
xy	1	1	1	1
$x\bar{y}$	1	0	1	1

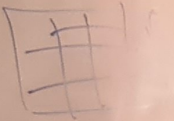
2) - Variables

Product of - Sum
Simplification

$$\bar{f} = \bar{x}v + \bar{y}\bar{z}v = (\bar{x} + \bar{v})(\bar{y} + \bar{z} + \bar{v})$$

	$\bar{z}\bar{v}$	$\bar{z}v$	zv	$z\bar{v}$
$\bar{x}\bar{y}$	1	0	0	1
$\bar{x}y$	1	0	0	1
xy	1	1	1	1
$x\bar{y}$	1	0	1	1

$$= (x + \bar{v})(y + \bar{z} + \bar{v})$$



Find the reduced form for the logic function
Presented in fig :

	$\bar{Z}\bar{V}$	$\bar{Z}V$	ZV	$Z\bar{V}$
$\bar{X}\bar{Y}$		1		
$\bar{X}Y$	1	1	1	1
XY	1	1	1	1
$X\bar{Y}$		1	1	